8.3 Box-and-Whisker Plots

Name _____ Period ____ Date _____

Algebra I 8.3 (30-minute plan) · Student sheet

I can…

- find the five-number summary of a data set.
- make and read a box-and-whisker plot.
- use the range, interquartile range, and shape to describe and compare data sets.

Vocabulary · fill in the missing word

Quartiles — values that _____ a data set into four equal parts: Q1 is the median of the lower half, Q3 the median of the upper half

Five-number summary — least value, Q1, median, Q3, _____ value

Box-and-whisker plot — a display of the _____ summary; the box goes from Q1 to Q3 with a line at the median, and the whiskers reach the least and greatest values

Interquartile range (IQR) — Q3 − Q1, the _____ of the middle half of the data

Part 1 · Notes and examples (fill in each note, then work the example)

Five-number summary: _____, _____, _____, _____, _____

Quartiles: Q1 is the median of the _____ half; Q3 is the median of the _____ half.

Example 1

Ages of 10 people at a party: 18, 25, 21, 30, 19, 23, 27, 22, 20, 35. Find the five-number summary and make a box-and-whisker plot.

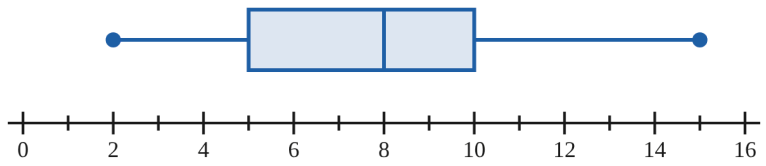
IQR = _____ − _____, the spread of the middle _____ of the data.

Example 2

Use the box-and-whisker plot.

a. Find the median. b. Find the range. c. Find the IQR.

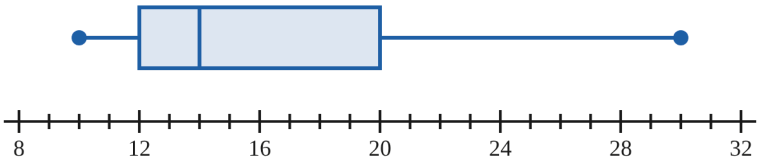
d. About what percent of the data are between 5 and 10?



Shape: A longer right whisker means the distribution is skewed _____.

Example 3

Describe the shape of the distribution.



Part 2 · Partner practice

Partner practice

Find the five-number summary and the IQR: 7, 15, 9, 12, 20, 11, 14, 8, 16